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The Compositionality of Concepts and Peirce's Pragmatic Logic

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1 Introduction

This paper discusses the contribution of the American philosopher and scientist Charles S. Peirce (1839–1914) to some questions concerning the compositionality of concepts and the role of context in logic.

Peirce developed the iconic, diagrammatic logic of *Existential Graphs* (EG) and provided semantics which earned the ahead-of-time but little-known title of the *Endoporeutic Principle* (EP) in 1905 (Pietarinen, 2004a).[1] This highlights the importance of *context construction* and *context updating* during interpretation. In conjunction with EGs, Peirce also emphasised strategic communication, mutual knowledge, and the presence of the *common ground* (Pietarinen, 2005b). His one-time student John Dewey (1859–1952) stated that the neglect of context is "the greatest single disaster which philosophic thinking can incur". It is the notion that is "inescapably present" without which we cannot "grasp the meaning of what is said in our language" (Dewey, 1931).

In addition to the role of context in logic, my purpose is to shed light on some of the ways in which EGs and the EP instantiate aspects of *compositionality* as well as *non-compositionality*, and to put these aspects into a historical perspective.

On the logical side, Peirce's pragmatist theory addresses precisely the Dewey contention. That is to say, there is no sentence, expression or depiction that has a meaning independent of the environment, context or circumstance within which it is uttered and within which it gets interpreted. This is not, of course, a spat against compositionality of meaning per se (apart perhaps from some very strong notions of compositionality in which the meaning of expression would be *exhaustively* defined by its parts), since context may itself be provided as a set

[1] Not all iconicity is diagrammatic: pictures and metaphors are also iconic representations, the logic of which is still being sought.

of utterances, or perhaps some representational schema, script, frame, or what have you, to which a linguistic or logical description can be attached. In other words, one may encode any contextual information into the model-theoretic consequence relation.[2]

Compositionality has often thought to be a prerequisite for learnability, systematicity, or even communicativity and comprehension of language (Pagin, 2003). For if proper subexpressions or subformulas are context dependent, it is said, they may not have a self-supporting meaning. It is for this reason, advocates of the compositional approach claim, that these expressions cannot be considered to be natural constituents of some larger unit, typically a formula or a sentence, the meaning of which ought to be morphically imaged on those constituents.

It is, however, by no means clear that such a claim for compositionality is valid from the historical point of view. I seek to substantiate this claim by investigating Peirce's pragmatic conception of logic deriving from the late 19th century.

2 Interpreting Existential Graphs

In Peirce's diagrammatic logic, graphs are used to assert things about individuals. The main components of such a system involve (1) *spots*, which are bounded regions of a surface, the *sheet of assertion* (the Phemic Sheet) upon which graph-instances are scribed, (2) closed, continuous, simple curves or *cuts* around a graph-instance representing negation of that instance, and (3) continuous, thick lines connected at the *hooks* on the peripheries of spots called *lines of identities*, representing existence, identity and predication. Spots represent unsaturated predicate terms Cuts are recursively nested. Identity lines are graph-instances, but their compositions, *ligatures*, are not.[3] The outermost free

[2]Encoding may be done directly into the sub-expressions of language as in *context-carrying* in logic programming or in *dynamic theories of meaning*, or else into extensions of assignment as in *compositional semantics* for 'Independence-Friendly' (IF) first-order logic, intended to provide proper semantic attributes via non-empty sets of sequences of assignments instead of merely sequences of them (Hintikka, 1996; Hodges, 1997; Janssen, 1997, 2002; Sandu & Hintikka, 2001).

[3]To be precise, Peirce held ligatures to be "composites" of several graph-instances (MS 669, 1911, *Assurance Through Reasoning*, available at http://www.helsinki.fi/˜pietarin/ with another unpublished manuscript with the same title, MS 670). The reference is to the Peirce Manuscripts (Peirce, 1967) by manuscript number, and, if applicable, followed by page number. If known, I also give the title and the year of the manuscript when first referred to.

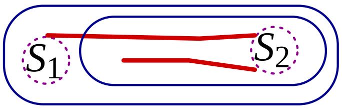


Figure 1: BETA graph with two cuts, two ligatures and two spots.

extremity of an identity line or ligature determines whether the quantification it represents is *universal* (the outermost free extremity lies within an *odd* number of cuts) or *existential* (the outermost free extremity lies within an *even* number of cuts).

We must skip over much of the details concerning EGs and refer to Pietarinen, 2004a,c,2005a,b; Roberts, 1973; Zeman, 1964). To see an example, Figure 1 depicts an EG that pertains to the BETA part of the system, in which the ligatures correspond to the existential ($\exists y$, denoted by the innermost extremity) and universal ($\forall x$, denoted by the outermost extremity) quantifiers of predicate logic. There is also a one-place spot S_1 and a two-place spot S_2.[4]

This graph may be correlated with the symbolic sentence $\forall x \exists y (S_1(x) \rightarrow S_2(x,y))$ of first-order logic. However, when doing things in this iconic and diagrammatic fashion, some characteristics are brough to the fore in iconic representations but not in symbolic expressions. Among them is continuity (see section 5).

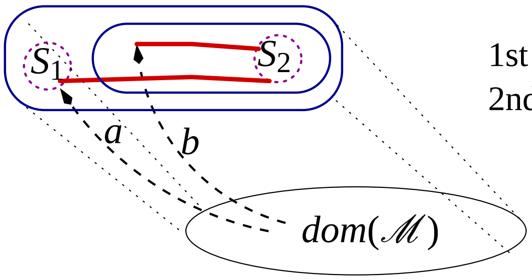
Graph-instances are interpreted according to the *Endoporeutic Principle* (EP): the outermost cuts (contexts) are evaluated on the model $\mathscr{M}$ before proceeding to the inner, contextually-constrained cuts. The evaluation of lines involves an instantiation of a value to the outermost end of an identity line, and this value then propagates continuously along the identity line towards the interiors of the inner cuts and to the spots to which the lines are attached. Peirce termed the process of instantiation, together with the type of the identity line, the *selective* (Figure 2). The EP thus provides the basic fashion in which contexts are created and updated.

As Figure 2 testifies, the evaluation takes place between two parties "in our make believe",[5] the GRAPHIST (alias the UTTERER) who scribes the graphs and proposes modifications to them, and the GRAPHEUS (alias the INTERPRETER) who authorises the modifications. We may think of them as players much in the sense of the *game-theoretic semantics* (Hilpinen, 1982; Hintikka, 1973; Pietarinen, 2003).

The graphs scribed by the GRAPHIST are true, because "the truth of the true

[4]If identity lines and hooks are omitted, one gets the ALPHA part of EGs, the theory of which is isomorphic to the theory of propositional logic.

[5]MS 280, c.1905, *The Basis of Pragmaticism*.



1st *selective*: the GRAPHEUS chooses a
2nd *selective*: the GRAPHIST chooses b

Figure 2: BETA graph, first two steps of the evaluation.

consists in his being satisfied with it" (MS 280: 29).[6] The GRAPHIST is therefore the verifier of the graphs. To end with a true atomic graph amounts to a win for the GRAPHIST, and to end with a false one amounts to a win for the GRAPHEUS. The truth of the whole graph agrees with the notion of the existence of a *winning strategy* for the GRAPHIST, in Peirce's terms a *habit* that is of "a tolerable stable nature" (MS 280: 30). Likewise, falsity is the existence of such habits for the GRAPHEUS.

Peirce assumes further that these players have *common knowledge* concerning the universe of discourse and thus a reasonable *common ground* well understood between the two of them, without which the discoursing that proceeds according to the presumption of collaborative enterprise would not be possible (Pietarinen, 2004e).

Summarising, a *semantic game* is played between the GRAPHIST and the GRAPHEUS on a given graph-instance G_i and a model $\mathscr{M}$, according to the following conventions:

1. *Juxtaposition* of graph-instances on positively (negatively) enclosed areas: the GRAPHEUS (resp., the GRAPHIST) chooses between two graphs. The evaluation proceeds with respect to that choice. The winning conventions change with respect to choices on negative areas.

2. Polarity of the area of the outermost extremity of a *ligature* determines whether the GRAPHIST (if positive area) or the GRAPHEUS (if negative area) is to make a choice from the $dom(\mathscr{M})$ to be the interpretation of the identity line. Evaluation proceeds with respect to that choice and with the graph-instance that has the ligature removed up to its connexions with spots. What will remain at the hooks of spots are zero-dimensional dots.[7]

[6]Note the resonance "being satisfied with" has with Tarski's somewhat less concrete idea of satisfaction. Also Löwenheim used a similar term of being "satisfied" ["erfüllt"] in his early model-theoretic explorations well before Tarski (Badesa, 2004: 139).

[7]Uninstantiated dots attached to hooks may be regarded as free variables of the ex-

3. When a *spot* is reached, its truth-value determines the winner: a spot that is satisfied in an interpretation means that the GRAPHIST wins a particular play on the graph, and the spot that is not satisfied in an interpretation means that the GRAPHEUS wins a particular play.

4. The existence of a *winning strategy* (i.e., a stable habit that leads to a win in all plays) for the GRAPHIST agrees with G_i being *true* in $\mathscr{M}$. The existence of a winning strategy (i.e., a stable habit that leads to a win in all plays) for the GRAPHEUS agrees with G_i being *false* in $\mathscr{M}$.

As regards to the EG in Figure 2, the evaluation commences with the selections of two individuals for the ligatures, the first by the GRAPHEUS and the second by the GRAPHIST. After that, the GRAPHEUS chooses between the spots S_1 and S_2 with a cut around it. If he chooses S_1, and S_1 is satisfied in $\mathscr{M}$, then the GRAPHEUS wins. If the GRAPHIST chooses S_1, and S_1 is not satisfied in $\mathscr{M}$, then she wins. The whole EG is true precisely in the case there exists a stable habit for the GRAPHIST, and false precisely in the case there exists a stable habit for the GRAPHEUS.

What are Peirce's views on the matters related to the compositionality of such graphs? In speaking about his EGs circa 1905, he made the following note in an unpublished draft:

> The *meaning* of any graph-instance is the meaning of the composite of all the propositions which that graph-instance would under all circumstances [= in all contexts] empower the interpreter to scribe. (MS 280: assorted pages 35).

A similar statement exists among other draft pages of the same manuscript:

> The *meaning* of any graph-instance is the meaning of the sum total or aggregate of all the propositions which that graph-instance enables the interpreter to scribe, over and above what he would have been able to scribe. (MS 280: assorted pages 35).

What the INTERPRETER (i.e., the GRAPHEUS) is empowered to scribe are thus, on the one hand, *experimental and evidenced facts* derived from experiments upon these diagrams, and on the other, *inferential propositions* that follow from graphs by the rules of transformation. According to Peirce, meaning therefore involves both inductive and deductive elements of reasoning.

Peirce's renowed *Pragmatic Maxim* (PM) says that the meaning of a concept is the sum total of its implications for possible observations and possible actions:

> The rule for attaining the third grade of clearness of apprehension is as follows: Consider what effects, that might conceivably have practical

tension of the system of EGs whose graph-instances would correspond to formulas.

> bearings, we conceive the object of our conception to have. Then, our conception of these effects is the whole of our conception of the object. (5.402, 1878, *How to Make our Ideas Clear*).[8]

This formulation of the PM first appeared in the January 1878 issue of *Popular Science Monthly*. Several versions of it exist in Peirce's large corpus. A very succinct and unambiguous one says that "the maxim of logic [is] that the meaning of a word lies in the use that is to be made of it" (CN 2.184, 2 February 1899, *Matter, Energy, Force and Work*).[9]

Given the PM, according to which the meaning of an assertion is the sum totality of all its actual and possible practical consequences under a given interpretation, a pragmatic principle of compositionality would be thus:

The Pragmatic Principle of Compositionality.The *meaning* of an assertion *A* is the meaning of all assertions that follow from *A* either by inductive or deductive principles and permissions under all authorised circumstances (i.e., those arising out of mutual consent by the GRAPHIST and the GRAPHEUS). Here we have an outward-looking, indefinitely-progressing principle for meaning. Moreover, in the light of such a pragmatic approach to logic, the *Context Principle* would be thus:[10]

The Pragmatic Principle of Context.A proposition has no meaning in isolation from its consequences. Accordingly, a proposition that has no consequences is meaningless.

3 The Endoporeutic Principle and Compositionality

In her exploration of the ALPHA and BETA parts of Peirce's EGs, Shin (2002) argues that one should dispense with the Endoporeutic Principle (EP). The claim Shin puts forward, namely that Peirce's obsession with the EP has foiled any comprehensive understanding of his system and its establishment in a wider perspective, is nevertheless mistaken. Her claim turns out to be a plea for compositionality, the notion of which we should not anachronistically attribute to Peirce. Shin does not refer to compositionality, but she laments that no real challenge has been voiced against the EP for reading the graphs. According to

[8] The reference is to Peirce (1931–58) by volume and paragraph number.

[9] The reference is to Peirce (1975–87) by volume and paragraph number.

[10] According to the usual formulation of the Context Principle, a word meaning cannot exist unless there is sentence in which words are embedded. Such views are found in Frege's (1884) *Grundlagen der Arithmetik*: "The meaning [*Bedeutung*] of a word must be asked for in the context of a proposition [*Satzzusammenhang*], not in separation" (cf. Beaney, 1997: 90).

her, the principle does not “reflect visually clear facts in the system”, and in fact “forces us to read a graph in only one way” (Shin, 2002: 63).

What are these allegedly visually clear facts? Shin refers to the impossibility of reading graphs so as to determine which are oddly enclosed and which are evenly enclosed by cuts. While the endoporeutic reading may deliver correct truth conditions for graphs, she thinks that the substitution of corresponding implications for nested cuts is often forced in an unnecessary way. Also, there is no mention in such a reading of a disjunction, namely the juxtapositions of sub-graphs encircled within an even number of cuts. Likewise, Shin claims, no vocal difference obtains between existential and universal quantification in the endoporeutic, outside-in reading.

To rectify these defects, Shin strives to reproduce the ALPHA and BETA parts in their respective equivalent negation-normal forms, namely those in which cuts are pushed in as deeply as possible. She terms the preferred reading a “direct reading” or a “multiple reading”, whereas the endoporeutic reading is “indirect”.

Stenning (2002) has employed similarly vague terminology in his discussion of the iconic nature of diagrammatic methods. Symptomatically, there is no mention of the dialogical and communicational interpretation of graphs in either Shin’s or Stenning’s works. I strongly doubt that a comprehensive overview of EGs can be obtained without such an interpretation. For instance, the difference between existential and universal quantification is precisely that which determines which of the parties, the GRAPHIST or the GRAPHEUS, is to select an instance from the universe of discourse as intended by the proposition that the graph depicts. The switching between them is then facilitated by the system of cuts. Likewise, the distinction between reading the graphs as representing conjunctions and reading them as representing disjunctions amounts to this role-playing view of game-theoretic semantics.

Shin’s remarks turn out to be a reflection of the wider and more far-reaching issues implicit in her discussion than ones merely related to the direction of reading the graphs. Although she does not spell it out explicitly, the absence of the two interlocutors in Shin’s discussion is characteristic of a more general presupposition, namely that of compositionality. It is this presupposition that lurks behind her intention to read iconic features of graphs in a way that she claims is not brought out by endoporeutic reading. The communicative and dialogical interpretation of EGs nevertheless reveals that such preoccupations are ill conceived. Given the EP, there would be nothing missing in the way EGs are read, since the distinction between different quantifiers and different logical connectives is exposed by the system of choices performed by the utterers and interpreters associated with the graph whose meaning is to be disclosed.

Another way of putting across a similar point is to note the equivocation in Shin's use of the term "endoporeutic" when compared to the way in which Peirce uses it. Shin speaks about "endoporeutic reading" of the graphs, while for Peirce, the EP was not a matter of reading the graphs, but a necessary consequence of the diagrammatisation that readily dictates how the graphs are interpreted. The EP provides a method for expressing the *truth* of the graphs, and only perhaps as a consequence, a method for reading them. Full understanding necessitates the two interlocutors, who choose and assign semantic values to each component in a definite order respecting the passage from outer instances to the inner, contextually-constrained instances. This is spelled out in an unambiguous manner in the game-theoretic interpretation, and is evoked in Peirce's notion of "nests" of graphs (4.472, c.1903; 4.494, c.1903; 4.617, 1908).[11]

Peirce noted the importance of the contextual and model-theoretic aspects encountered by those who interpret EGs:

> The rule that the interpretation of a graph must be endoporeutic, that is, that the graph of the *place* of a cut must be understood to be the subject or condition of the graph of its *area*, is clearly a necessary consequence of the fundamental idea that the Phemic Sheet itself represents the Universe, or primal subject of all the discourse. (MS 500: 48, 1911, *A Diagrammatic Syntax*).

The usefulness of this interpretation is beyond dispute in the interpretation and understanding of *anaphoric* expressions (and not only nominal anaphora but also those involving temporal co-references). In comprehending discourse, what other way is there to bring the values of anaphoric pronouns into being than by looking at what has happened in previous rounds of interpretation, interspersed with the contextual and environmental input that the Phemic Sheet, namely all that is well understood between the utterer and the interpreter, is supposed to provide?

According to Peirce, "Endoporeusis, or inward-going" (MS 650: 18), is like a global clock that synchronises interpretation and arranges it in an outside-in fashion. Metaphorically, it is like:

> a march to a band of music, where every other step only is regulated by the arsis or beat of the music, while the alternate steps go on of themselves. For it is only the iteration into an evenly-enclosed area that depends upon the outer occurrence of the iterated graph, the iteration

[11] "The interpretation of existential graphs is endoporeutic, that is proceeds inwardly; so that a nest sucks the meaning from without inwards unto its centre, as a sponge absorbs water" (MS 650: 18, 1910, *Diversions of Definitions*). The "nest" is here a technical term meaning the sequence of cuts from those that have the largest number of enclosures to those that have the fewest.

> into an oddly enclosed area being justified by your right to insert whatever graph you please into such an area, without being strengthened or confirmed in the least by the previous occurrence of the graph on an evenly-enclosed area. So the analogy to a march is pretty close. (MS 650: 18–19).

The question for Peirce was therefore not how to read off what the graphs are intended to convey. There are numerous logically equivalent ways of doing so. The question was: What is the most appropriate "system for expressing truth endoporeutically" (MS 650: 19)? This is what the pragmatist has in view, namely a definite purpose in investigating logical questions, since "he wishes to ascertain the general conditions of truth" (2.379, 1901, *Negation*). Peirce predicted that "if anybody were to find fault with the system ... I should be disposed to admit that it is a poetical fault" (MS 650: 19).

In pure aesthetic terms, there may well be alternatives, but such alternatives ought to achieve at least as simple a set of rules as endoporeusis does, and in addition predict the same merits for its essential features. Alternatives to EP have not fulfilled these predictions. Shin's proposal to reduce the ALPHA graphs to the negation-normal form involves a recursive definition that starts from atomic graphs outwards. Proof-theoretically, replacing Peirce's five transformation rules of erasure, insertion, iteration, deiteration and double cut, Shin's reformulated inference rules come in seven parts, four of which are partial decompositions of the rules of iteration and deiteration into the rules that have been used in natural deduction systems. In the case of the BETA part, there are five rules for their transformation, which Shin rewrites into nine rules divided into twenty-six sub-cases.

In general, then, compositional methods of interpretation are far more complex in that they are liable to have a considerably larger number of transformation rules than those that Peirce proposed.

Furthermore, extensions of EGs, such as adding more dimensions into the sheet of assertion (Pietarinen, 2004c) is bound to increase the complexity of any compositional alternative to the EP to an unnatural extent.

In stark contrast to the animadversions professed in Shin's account of EGs, Peirce did not possess comparable predispositions concerning the preferred interpretation of natural language. For him, the reason was historical rather than systematic. His sin was not that he overlooked the distinction between what is compositional and what is non-compositional in logic, since no such distinction was forthcoming in EGs. Nor was it to remain preoccupied with only one, the endoporeutic, semantics of his graphs, since this was the only context-abiding game of interpretation in town. His sin – if indeed it was such – was simply to fix his ideas by presenting a multiplicity of diagrammatic systems that far exceeded what was comprehensible (and printable) at that time.

4 The Composition of Concepts

In discussing context and compositionality, I alluded to the contributions by diagrammatic, iconic approaches to logic. These approaches are rich both in content and historical significance. What is achieved via representations such as EGs has not been brought to bear on compositionality before. The acute nature of such a discussion is shown by the well-argued similarities between, on the one hand, the thought in its relation to the mind, and on the other, iconic representations as ‘snapshots’ or images of the essence of the mind in action. I will turn to this topic next.

In their argumentation concerning the status of compositionality, Fodor & Lepore (2002) started with the premise that the content of a sentence is the thought or proposition it is used to express. I will leave it to others to decide whether substantial issues exist which prohibit equating content with meaning. In the light of the PM, this premise is nevertheless clearly inept. Thoughts, let alone propositions or propositional contents, have little to do with the characterization of meaning which ought to be intersubjectively and cognitively testable, verifiable, or observable, or, in the very least, potentially or subjunctively so.

What is more, Peirce’s logic and his EGs exhibit some direct links with cognitive issues. I will adduce evidence for the view that Peirce did recognise the significance of how concepts are composed and the relationship between the logic of EGs and the prospects of achieving a compositional mapping between a representational system and the mind.

Peirce demonstrates the reality of such links by stating that, just as thoughts are *determinations* of the mind, graph-instances are *determinations* of the blank sheet of assertion.[12] The mind is a composition of thoughts in a manner analogous to that in which graphs are compositions of their graph-instances. The mind is a comprehensive thought in a manner analogous to that in which the sheet of assertion, with all its permissible transformations and actual transformation-instances, is a well-formed graph.

The second main interconnecting feature is that graph transformations and identities are *continuous*, and so is the thought that a graph, together with all its practical consequences, represents. I return to this feature in the next section.

Unfortunately, some crucial passages, including parts of manuscripts 498 and 499 in which Peirce promises to tackle the nature of the *composition of concepts* and the nature of the proposition through EGs, have not been recovered among his vast corpus of unpublished manuscripts.[13] It is likely that these drafts were

[12]MS 490, 1906, *Introduction to Existential Graphs and an Improvement on the Gamma Graphs*.

[13]These drafts were prepared as preliminary addresses to one of the 1906 meetings

left incomplete and Peirce never brought the promised issues to a conclusion. In MS 498 he notes that:

> The solid core of all this discussion is the question of how concepts can be compounded. Suppose two concepts, A and B, to be combined. What unites them? There must be some cement; and this must itself be a concept C. So then, the compound concept is no AB but ACB. Hereupon, obviously arises the question how C is combined with A or with B. The difficulty is obvious, and one might well be tempted to suspect that compound concepts were impossible, if we had not the wish manifest evidence of their existence.
>
> Here, then, are the two puzzles of logic upon which I am going [to] try what light can be shed by the system of Existential Graphs. They are the puzzle of the relation of signs to minds, and of their communication from one mind to another, and the puzzle of the composition of concepts and the nature of the judgment, or, as we of the antipsychological school say, of the proposition.

The rest of the manuscript provides a preparatory discussion of the system of EGs. In the slightly later MS 499, he also starts in a promising way:

> Another great puzzle of logic is that of the composition of concepts, or thoughts. It is evident that a thought may be complex. ... there are thoughts compounded of thoughts. But let A and B be two simple thoughts which can be compounded. But how are they compounded? They are compounded in thought. Very well, then, the composition must then be a third thought, which we may denote by C; so that the compound is not AB but is ACB. Then the question arises, how are A and C compounded, as they certainly are in ACB; and how are C and B compounded? This puzzle has been formulated in all its generality by some great logicians.

There is then a lengthy intermission concerning the nature of assertions and judgements. A couple of pages further on, Peirce characteristically admits that:

> It is still unsettled and is the most prominent perhaps among unsettled questions in the logical literature of recent years. I wish in this communication to exhibit to you the unexpected solution not merely of this problem but of the more general problem of the composition of concepts to which the systems of existential graphs leads.

of the National Academy of Sciences. They are entitled *On Existential Graphs as an Instrument of Logical Research* (MS 498) and *On the System of Existential Graphs Considered as an Instrument for the Investigation of Logic* (MS 499). MS 490 is the version Peirce ended up presenting at April 1906 meeting in Washington DC.

> I do not propose today to enter into the demonstration of the truth of the solutions suggested by existential graphs, of the two problems in regard to the relation of signs to the mind and of the composition of concepts. The reasons to exclude this part of the discussion are, first, that it would render my paper tedious since the proof presents no very striking idea, or other great novelty, and second, that if I should go into the tedious development it is unlikely that any one member would carry it away from an oral statement without dropping out some point which is essential to its cogency; and at any rate it would be unintelligible to the great majority. In case the paper should be printed, I will append the proof for the benefit of those who may desire to examine it.
>
> I have now sufficiently indicated two great logical puzzles. I do not call them problems because it is the nature of logical difficulties that until they are solved we cannot distinctly state what the problem is. Even after they are solved, it is often no easy matter to say what the problem was. The puzzle of the measure of force is one of [many] instances of this that I might adduce. These two puzzles relate to the mode of composition of concepts in general and to the nature of the proposition in particular, and to the relation of concepts and signs to the mind.

Following this, Peirce goes on to discuss some general issues concerning the system of EGs.

Luckily, however, when addressing related issues in 4.572, Peirce briefly notes how EGs are intended to solve the puzzle concerning the composition of concepts. The problem is that:

> the ways in which Terms and Arguments can be compounded cannot differ greatly from the ways in which Propositions can be compounded. A mystery, or paradox, has always overhung the question of the Composition of Concepts. Namely, if two concepts, A and B, are to be compounded, their composition would seem to be necessarily a third ingredient, Concept C, and the same difficulty will arise as to the Composition of A and C.[14]

The solution to this is, in Peirce's own words, the following:

> As far as propositions go, and it must evidently be the same with Terms and Arguments, there is but one general way in which their Composition can possibly take place; namely, each component must be indeterminate in some respect or another; and in their composition each determines the other.

[14]From *Prolegomena to an Apology for Pragmaticism* published in *The Monist* in 1906, submitted in April that year.

Speaking extensionally (viz., 'on the *recto*' of the EGs), this is effectuated such that the indefinite sentence

> "Some man is rich" is composed of "Something is a man" and "something is rich," and the two somethings merely explain each other's vagueness in a measure. Two simultaneous independent assertions are still connected in the same manner; for each is in itself vague as to the Universe or the "Province" in which its truth lies, and the two somewhat define each other in this respect.[15]

The interconnectedness of signs on the sheet is thus but one manifestation of Peirce's *synechist* doctrine of continuous connections and mutual attraction between signs (representamens) that the mind composes in order to form wholes and thus becoming capable of comprehending and interpreting.

As far as the conditional is concerned, a similar interplay between the concepts of the antecedent and the consequent is manifest:

> The composition of a Conditional Proposition is to be explained in the same way. The Antecedent is a Sign which is Indefinite as to its Interpretant; the Consequent is a Sign which is Indefinite as to its Object. They supply each the other's lack. (4.572)

In a similar vein, in parts of manuscript 490 that were omitted from its publication in 4.582, Peirce notes that graphs are compounded and united by combinations of "being the relate and the correlate of a relation". This natural union of the relate and the correlate as a special determination of the sheet of assertion, the union signifying existential relations, verb actions and identities alike, was the diagrammatic counterpart of the special determination of the mind by an idea.[16]

5 Continuity

To provide an example of the application of Peirce's synechist outlook on logic, the following remark can be made. It has been one of the grand mysteries

[15] See also 4.561fn, which is from MS 300: 33–39, the intended follow-up to the *Prolegomena* series entitled *The Bed-Rock Beneath Pragmaticism*, written in 1905. Cf. 499 (supplement): "Existential Graphs shew beyond all doubt to the discerning mind that the Composition of Concepts can only take place by the reciprocal precisions of indefiniteness".

[16] Furthermore, it was this union of the relate and the correlate that gave rise to the difficulties of any straightforward composition with respect to modal and actual universes and to the puzzles concerning possible individuals, subsequently part of 'cross-world identification' of quantified modal logics (Pietarinen, 2005a).

in his later logical writings as to what he may have meant in speaking about *continuous predicates* in relation to his EGs. For instance, at an age of 69, he wrote to Victoria Welby the following:

> A predicate which can thus be analyzed into parts all homogeneous with the whole I call a *continuous predicate*. It is very important in logical analysis, because a continuous predicate obviously cannot be a *compound* except of continuous predicates, and thus when we have carried analysis so far as to leave only a continuous predicate, we have carried it to its ultimate elements. I wont lengthen this letter by easily furnished examples of the great utility of this rule. (SS: 72, 1908, *Letter to Lady Welby*).[17]

As far as I have been able to see, no explanation of what Peirce could have meant by continuous predicates is forthcoming elsewhere in his writings. However, an enlightening possibility here is to think of predicates p as subsets A of X in the mappings from a space X to the set $\{0,1\}$, where X is an arbitrary topological space. Continuity of the predicate would now depend on the topology defined for $\{0,1\}$. Moreover, continuous functions to a two-element lattice now correspond to open sets of X. Hence, a relationship has been drawn between predicates and open sets. This is, in fact and in brief, the formal way to derive continuous predicates. Now, is this something that Peirce could have envisioned? The answer is positive, because as Peirce stated in the above quotation, a continuous predicate is also “a predicate which can thus be analyzed into parts all homogeneous with the whole”. This is but an alternative way of saying that a continuous predicate can be characterised as the set of all predicates consistent with it.

Moreover, the iconic, or diagrammatic, method of representing assertions and reasoning about them is, as Peirce noted, best conducted by establishing logical principles upon topological, continuous principles. The state of these principles was quite elementary at that time, but also quite rich in the sense of not being preoccupied with the later developments concerning set theory and the nature of continuum during the early phases of the symbolic era in logic (Pietarinen, 2005a).

The explication of continuous predicates may seem somewhat cursory or overly specialised for the main purposes here, but I believe that it illustrates the more general and serious concern this early contribution to cognitive science and logic raised than the technicalities indicate. Namely, the general point is that semantics can and should be based on principles that have a continuous core. This may happen either in the semantic attributes or in describing the

[17] The reference is to Peirce (1977) by page number.

parts of the representations (say, in iconic, diagrammatic or metaphoric mental images) as being continuously connected ('path-connected') with one another.

To substantiate the associations, we may allude to works such as Wolfgang Köhler's *Field Semantics* for the theory of perception dating back to the 1920s, namely that "neural functions and processes with which the perceptual facts are associated ... are located in a continuous medium" (Köhler, 1940: 55). Notable is also Peirce's remark in his statement quoted above that spots, propositions and inferences are to be continuously connected with one another. Topologically, this expresses connectivity between different parts of the surface, thus forming propositions, and in terms of propositions being connected, in the equally topological sense, under continuous transformations from one graph to another, thus forming inferential arguments.

By itself, the conceptualisation of a continuous predicate and more generally the continuous connections between separate parts of the representations already bore the traces of the later Gestalt or perceptive schemes rather than anticipating anything like the Tarski-type logical semantics which have an inherently compositional and recursive nucleus.

However, Peirce did not make the connection with the later truth-conditional and semantic issues explicit, as he was first and foremost striving to account for what would count as necessary reasoning in terms of the dynamic understanding of 'motion', for instance through continuous morphing from one diagram onto another according to given permissions (Pietarinen, 2005b). Hence, these wider concerns were easily sidestepped by the advocates of discrete-symbolic logic who started to gain ground shortly after the forgetfulness of these early contributions commenced.

6 Dynamics and Discourse

The aforementioned perspectives on Peirce's pragmatic logic also highlight what is misconceived in recent dynamic theories of meaning. They have attempted to characterise central semantic notions in terms of context-change potential (van Benthem, Muskens & Visser, 1997). What these formalisms have been seeking, though, is not meaning in the pragmatic sense of a list of practical consequences, but a transfer between varying ways of assigning interpretations to structures within which expressions are evaluated. It is effected by input-output pairs of assignments to a formula.[18]

Note how smoothly this links with the agenda laid out in the introduction (note 2), in which I implicitly called for proper motivation for providing the semantic entities in some of the recent formulations of compositional semantics.

[18]By doing this, dynamic semantics has come to characterise a limited form of *relevance* for expressions in terms of context change, but certainly not in terms of their meaning.

Like dynamic semantics, these compositional semantics have accomplished little more than an extension of the concept of assignment. For instance, in building compositional semantics for logics of imperfect information (a.k.a. IF logic) the extension is from sequences of assignments to sets of sequences of assignments. In dynamic semantics, the extension is from sequences of assignments to pairs of sequences of assignments. Neither case explains what the relevance of the excessive semantic material smuggled in by the extensions of assignments amounts to in the course of an interpretation. Surely not all meaning lies in the assignments.[19]

The centrality of context is cogently emphasised in comparing the EP of EGs with that of dynamic semantics. Given the iconic and topological nature of the method by which graphs are scribed on the assertion sheets, the logic is readily and inherently dynamic. For consider the EGs with two spots and an identity line that has one free extremity in Figure 3. These three graphs are equivalent. But they do not make a similar distinction as symbolic sentences that have two different accounts of the scope of the existential quantifier, namely $\exists x\,(S_1(x) \wedge S_2(x))$ and $\exists x\, S_1(x) \wedge S_2(x)$.

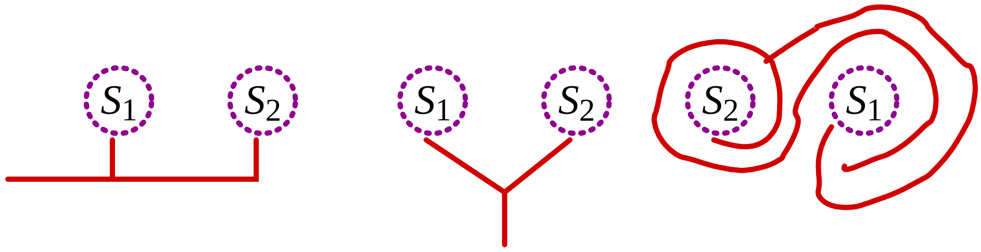


Figure 3: Three equivalent existential BETA graphs.

Accordingly, the logic of EGs is dynamic in that is does not differentiate between, on the one hand, $S_2(x)$ within the *syntactic scope* of the initial existential quantifier but not within its *semantic scope* as in $\exists x\, S_1(x) \wedge S_2(x)$, and, on the other hand, $S_2(x)$ being within both the syntactic and semantic scopes of the quantifier as in $\exists x\,(S_1(x) \wedge S_2(x))$. As soon as the system of lines of identities, the ligature, extends to both of the spots juxtaposed on the sheet and branches to attach to the hooks on the peripheries of both S_1 and S_2, the spots become semantically bound by the line. Such a line always has an interpretation. Since the spots are thus assigned a semantic value, there is no fear of any violation of the compositionality of the truth of these graphs in the sense of including

[19] What the relevance-theoretic framework of Sperber & Wilson (1995) attempts is to accept only those semantic entities that are context-effective in the cognitive, strategic and collateral senses, in other words items that have some tangible, realistic and pragmatic role to play in future cycles of interpretation.

components that lack an interpretation.

On the other hand, conjunction as juxtaposition is commutative, which nevertheless is not the case in dynamic semantics. However, similar dynamism obtains just as well with connectives other than conjunction. For instance, we may have an EG that represents a dynamic conditional, which Peirce termed iconically the *scroll*, in other words the continuous line of a cut folded in on itself. The scroll shows how information on the outer interior of the cut is transmitted to the inner interior of the cut. (This was used already in Figure 1.) Figure 4 now depicts three equivalent EGs in a fashion similar to those in Figure 3.

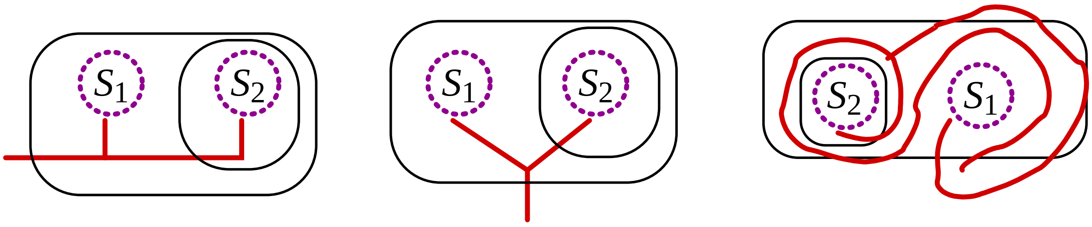


Figure 4: Three equivalent existential BETA graphs.

These three graph-instances are all equivalent in that, as soon as the ligature extends from the outer area of the scroll to the inner area, the corresponding spots become bound by it, and accordingly are saturated by the values selected for the identity line in question.[20]

The selection of values and their instantiation in identity lines (i.e., the *selectives*) is crucial. As noted, such a selection is performed by evoking two make-believe parties, the GRAPHEUS and the GRAPHIST, both of whom have a specific, truth-conditional objective in view. Purpose is, of course, also at the heart of pragmatic logic itself. Peirce elaborated this by saying that the pragmatist has in view a "definite purpose in investigating logical questions", and that he "wishes to ascertain the general conditions of truth" (2.379).

The parties are assumed to have acquired collateral information from observation and experience. It is thus the purpose, together with the common knowledge that constitutes their common ground, including the memory of their own factual and conceptual knowledge, which enables them to check for relevance and to make correct predictions (Pietarinen, 2004b). But they also update that knowledge and the common ground as interpretation proceeds, since in encountering logical constants while evaluating the graphs according to the EP, the val-

[20]It is a measure of the generality of diagrammatisation that pronouns may, with no extra effort, be also anaphorically bound to a universal quantifier (*Every player picks a card. He puts it in his hand*). This has proved to be beyond the expressive power of standard dynamic semantics.

ues chosen for them are added to their stock of information. By doing this, the parties also come to a mutual realisation of the general conditions concerning the truth of the assertions that they undertook to discourse upon. It is precisely in this manner that the logic of EGs fulfils the pragmatist agenda that Peirce set out.

It is also in these pragmatistic mindsets that the logic of EGs and their endoporeutic interpretation is at the same time both *verificationist* (in a manner and extent similar to that in which the pragmatic maxim is verificationist) and *truth-conditional*, taking the best of both worlds.[21]

John Sowa (1997) has argued that EGs are isomorphic to the discourse-representation structures (DRS) of the *discourse-representation theory* (DRT; Kamp, 1981; Kamp & Reyle, 1993). DRT is nevertheless non-compositional in that no algorithm exists that would reproduce the structure of natural language in a compositional way at the level of DRSs. Dynamic semantics was created to make the processing compositional by eliminating the level of DRSs and replacing it with the language of symbolic logic (Groenendijk & Stokhof, 1991, 2000). The non-compositionality of DRT appears, according to this view, to be on a par with that of the non-compositionality of EGs. However, I also noticed that EGs are similar to dynamic theories of logic in relaxing the notion of the scope of quantifiers, and in distinguishing between binding a variable and logical connectives being syntactically subordinated in terms of adhering to linear, step-by-step interpretation.

Far from exhibiting any tension, this affinity of EGs to both DRT and dynamic semantics only serves to vindicate the versatility of setting up logical systems in an iconic, diagrammatic manner. It would be a *non-sequitur* from this to expect a thoroughly compositional process to run all the way from natural language to its interpretation, possibly via some intermediate iconic representation, since compositionality is understood quite differently in diagrams. The sense in which EGs may be rendered compositional is to take the dynamic approaches that extend the concept of the assignment as carrying over to the corresponding extensions of the semantics of EGs, starting with Tarski-type semantics for the existential BETA graphs (Burch, 1997; Hammer, 1998).[22] Without such an extension, the EP of EGs is indeed non-compositional in the broad sense in which DRT is non-compositional, namely representing natural language in more than one dimension and then utilising an outside-in interpretation of instantiation of selectives. Equivocation in compositionality is imminent when moving from

[21] Accordingly, arguments for game-theoretic semantics (Hintikka, 1987) also hold for EGs.

[22] These authors fail to note that their semantics should not distinguish between the three graphs depicted in Figures 3 or 4.

discrete-symbolic representations to continuous-iconic ones.

7 Conclusions

The question of the relative priority of compositional and non-compositional systems was something that simply did not present itself as the most-central issue in Peirce's logical investigations or, in broad terms, in his semeiotics. In setting up EGs, several issues were encountered which, only in retrospect, pertain more or less intimately to issues concerning the compositionality of interpretation.

While Peirce made several choices that ultimately led to compositional meaning, it is unlikely that these choices were motivated by considerations featuring in contemporary literature, including those of learnability, systematicity or communicativity of language. The most important choice of all, the Endoporeutic Principle, erupted upon him with such cataclysmal force that it remained the one and the only game in town that he ever came to consider and endorse.[23]

That Peirce would have been keen to repudiate the utility of any alternative compositional version of semantics over his non-compositional one is supported not only by his overall pragmatic and exterior-to-interior method of interpretation, but also by his ever-present *holistic* attitude to recognition, perception, and other cognitive tasks.

Such an Aristotelian *holon* takes graph-instances, definite but incomplete snapshots of the contents of the mind, as a unity regulated by the true continuity in synechism, mathematically rendered as topological, truth-preserving mappings from one graph-instance to another. If, as a result of violating transformation rules, the relative positions of graph-instances on a given graph change, this change must affect the composition of the given graph as a whole. Analogously, if the consequences that follow from the graph and which constitute the aggregate of meaning as the collection of graph-instances change, this change must affect the nature of the meaning of the given graph.

In doing this, Peirce came to anticipate the later Gestalt theories of representation and Gestalt-like organisation and the "mental composite photograph"[24] type of composition of percepts (Pietarinen, 2004d). He also anticipated the later non-monotonic logics via his *tychims* (Marostica, 1997), a doctrine of absolute chance which, synthesised with pragmatism, gave rise to early excursions

[23]The reviewer notes the paper of Pagin & Westerståhl (1993), which presents a flexible outside-in binding of variables for predicate logic to account for natural language anaphora, and which has a postscript in which the authors briefly compare their system with Zeman's (1964) rendering of Peirce's EGs.

[24]2.438, c.1893, *The Grammatical Theory of Judgment and Inference*.

in logically exact accounts of continuity.

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